IE 5374: Foundations for Data Analytics Engineering

Prof. Mohammad Dehghani

"In God we trust. All others must bring data."
- W. Edwards Deming

Foundations for Data Analytics Engineering

IE5374

Course Info

- Prereq: None
- Tue & Thurs
- 09:00am - 12:20pm
- Room 119, Dodge Hall

Instructor Info

- Mohammad Dehghani
- Office Hrs: Mon 9:00am - 10:00am
- Zoom (make an appointment)
- Website
- m.dehghani@northeastern.edu

TA Info

- Asif Khan
- Office Hrs: Mon & Fri 10:00am-12:00pm
- Teams (make an appointment)
- khan.asif@northeastern.edu
- Shah Zeb (Kazi)
- Office Hrs: Mon & Fri 10:00am-12:00pm
- Teams (make an appointment)
- zeb.s@northeastern.edu

Course Overview

IE 5374 offers students an opportunity to learn fundamentals of computation required to understand and analyze real world data. The course helps students to work with modern data structures, apply data cleaning and data wrangling operations. The course covers conceptual and practical applications of probability and distribution, cluster analysis, text analysis and time series analysis.

Course Objectives

- Learn to work with modern data structures.
- Learn to create, clean, organize, and manipulate multidimensional data.
- Learn to apply concepts of probability and distributions for analytics.
- Learn to perform matrix operations
- Learn to build procedures for cluster analysis
- Learn to apply computational techniques for text analysis
- Learn to conduct descriptive and predictive time series analysis

Reference Books

- Main Textbook
  - An Introduction to Statistical Learning Website
  - Data wrangling with R Snell Library
- Additional References (optional)
  - Walpole, R. E., Myers, & R. H. (2012). Probability & statistics for engineers & scientists. Pearson Education Limited.
  - David Hand, Heikki Mannila, Padhraic Smyth (2001). Principles of Data Mining, The MIT Press, ISBN-10: 026208290X, ISBN-13: 978-0262082907.
  - Chatfield, C., & Xing, H. (2019). The analysis of time series: an introduction with R. CRC press.
  - Silge, J., & Robinson, D. (2017). Text mining with R: A tidy approach. " O'Reilly Media, Inc."
  - There are also plenty of online resources for R. Here is a Twitter feed of posts by Rbloggers R Bloggers

Outline

- Modern data structures, control structures and conditional statements
- Data cleaning, data wrangling
- Custom functions
- Probability
- Conditional probability and bayes rule
- Random variables, mean, variance, co-variance
- Discrete and continuous probability distributions
- Joint, marginal distributions
- Conditional probability
- Eigenvalues and eigenvectors, positive semidefinite matrices
- Cluster analysis
- Text analysis
- Time series analysis

Class Schedule

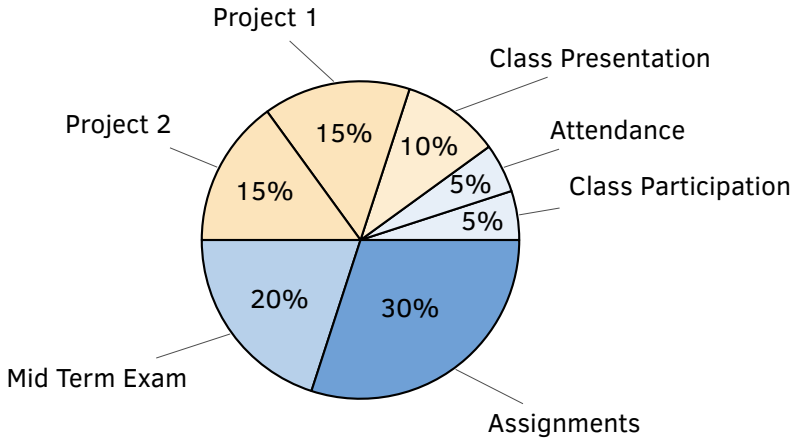
Please check Canvas for the weekly schedule of the course. If due to unforeseen circumstances the schedule changes, students will be informed prior to the change in schedule.

Software Tools

Software R is required for this course. R is a freely available language and environment for statistical computing and graphics. You must have access to a computer with R to be able to complete this course. Installing both R and RStudio on your own computer is the best option. RStudio is not strictly required, but several R concepts will be introduced with RStudio in mind, thus it is highly recommended.

- Installation guide: Below are the instructions to install R and R Studio:
  - : Follow the instructions in the link below to install R and R Studio for Mac. All the steps remain the same as shown in the video except the R and R studio version. Latest version is R 4.1.0 and R Studio 1.4.1713 (Video)
  - : Follow the instructions in the link below to install R and R Studio for Windows. All the steps remain the same as shown in the video except the R and R studio version. Latest version is R 4.1.0 and R Studio 1.4.1713 (Video)

Grading Scheme



- Assignments (Individual)
  - Assignments will be released on a weekly basis.
  - All homework shall be submitted through appropriate mediums as communicated by the professor/TAs
  - Homework and labs submitted have to be labeled as "IE 5374 - HW # (or Lab #) - Sec # - <Student Name>"
  - Late assignments *will not be* graded.
- Labs (Individual)
  - In-depth lessons are followed by R labs to help students develop their programming and statistical analysis skills.
  - Each lab consists two files, (i) lab document (*.pdf), and (ii) code file (*.R).
  - Students are *required* to review the lab PDF file prior to lectures to be familiar with the lab contents.
- Exams: Midterm Exam (Individual)
  - Midterm Exam will be given after covering the first 5 chapters.
- Case Study (Group based)
  - This part are required to form groups of 2-3 students.
  - The case study requirements and deliverable will be posted on CANVAS.
- Class Participation & Discussion
  - The course is synchronous and attendance is mandatory for students taking the course online or on-ground. To have an effective learning experience, students are encouraged to have continuous interaction with the faculty. A discussion question will be given to each section/break-out rooms to stimulate critical thinking. All students need to actively participate in discussions to articulate their ideas and share their thoughts.
  - Performance will be evaluated based on their participation and quality of response.

Your grades will be posted on CANVAS. You can check your running grade at any time, so there are no surprises. In fact, you should audit the correct entry of your grades. No grade corrections will be made after the last week of the semester. Final Grades and Course Letter Grades are not curved and are assigned as follows:

Letter Grade	Points	Letter Grade	Points	Letter Grade	Points
A	93-100	B	83-86	C	73-76
A-	90-92	B-	80-82	C-	70-72
B+	87-89	C+	77-79	F	<70

Emailing Us?

Please use the following format for the subject line of your email to avoid miss communication. This subject line will reduce the confusion.

IE5374 - Sec # - Subject Line

Make-up Policy

- No makeup opportunity will be given for any homeworks/lab assignment/projects.
- Students unable to attend classes are responsible for obtaining information regarding announcements and lecture content

Class Policy

- Students are expected to attend all class sessions and are responsible for all materials presented in class. Past experience has shown students who attend regularly and interact with instructors/classmates tend to be better overall.
- Laptops: You may need to use your laptop during the class upon the instructor's notes
- Phones/tablets/smart watches: Electronic devices may not be used during the class. If you have class work, social networking or sports-score browsing, that is more important to do, then please skip class. Life is full of such choices.
- Discussion between students for homeworks and projects is permissible. However the ideas / solutions / codes must not be copied. Each individual submission must be an outcome of individual effort

Classroom Recordings

This course, or parts of this course, may be recorded for educational purposes. These recordings will be made available only to students enrolled in the course, instructor of record, and any teaching assistants assigned to the course.

Teaching Modality

The university is transitioning into a fully in-person Fall 2021 term with all campus activities resuming. I will be teaching in person, on campus, and in the classroom. Students are expected to be in the classroom, no longer alternating between remote and in-person learning. Classroom density restrictions have been lifted, and we will no longer be using the Dynamic Scheduler. However, some students may need to be absent from the classroom due to illness, travel/visa issues, or are waiting for a negative test at the start of the fall term before attending their first class. Beyond students already approved for remote learning, all others must notify faculty (i.e., me) of any of the aforementioned reasons for missing class an in-person in order to be accommodated.

Academic Integrity

A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University.

As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, examinations of whatever kind, collaborative work, the use of study aids, the appropriateness of assistance, and other issues. Students are responsible for learning the conventions of documentation and acknowledgment of sources in their fields. Northeastern University expects students to complete all examinations, tests, papers, creative projects, and assignments of any kind according to the highest ethical standards, as set forth either explicitly or implicitly in this Code or by the direction of instructors.

Go to Academic Integrity website to access the full academic integrity policy.

Student accommodations

Northeastern University and the Disability Resource Center (DRC) are committed to providing disability services that enable students who qualify under Section 504 of the Rehabilitation Act and the Americans with Disabilities Act Amendments Act (ADAAA) to participate fully in the activities of the university. To receive accommodations through the DRC, students must provide appropriate documentation that demonstrates a current substantially limiting disability.

For more information, visit Disability Resource Center website

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Diversity and Inclusion

Northeastern University is committed to equal opportunity, affirmative action, diversity and social justice while building a climate of inclusion on and beyond campus. In the classroom, members of the University community work to cultivate an inclusive environment that denounces discrimination through innovation, collaboration and an awareness of global perspectives on social justice. It is my intention that students from all backgrounds and perspectives will be well served by this course, and that the diversity that students bring to this class will be viewed as an asset. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, socioeconomic background, family education level, ability - and other visible and nonvisible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. Your suggestions are encouraged and appreciated.

Please visit the website for complete information on Diversity and Inclusion.

TITLE IX

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.
Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty and staff.
In case of an emergency, please call 911.

Please visit the website for a complete list of reporting options and resources both on- and off-campus